



JEE(Advanced) : MOCK TEST

PAPER-1

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

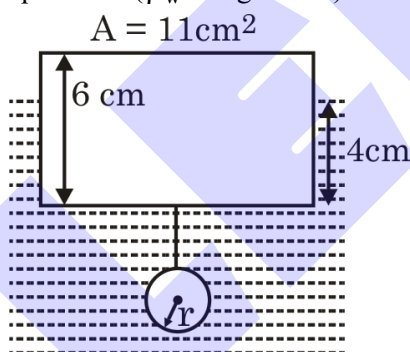
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

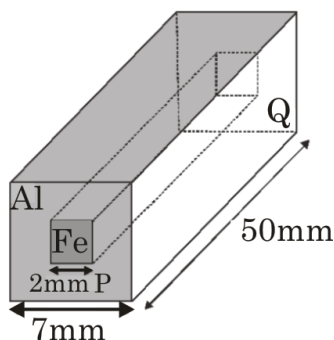
Negative Marks : -1 In all other cases

1. Figure shows a metal ball suspended by thread of negligible mass from an upright cylinder that floats partially submerged in water. The cylinder has height 6 cm, face area 11 cm^2 on the top and bottom and density 0.5 g/cm^3 . 4 cm of cylinder's height is inside the water surface. If density of the metal ball is 8 g/cm^3 then its radius is equal to :- ($\rho_w = 1 \text{ g/cm}^3$)

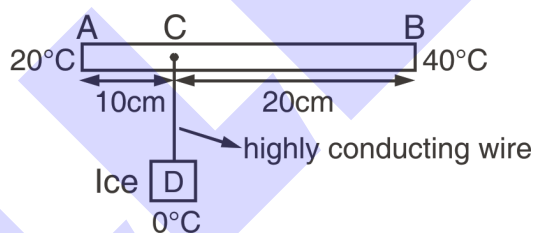


- (A) $\left(\frac{3}{8}\right)^{1/3} \text{ cm}$ (B) $\left(\frac{3}{4}\right)^{1/3} \text{ cm}$
- (C) $\left(\frac{4}{3}\right)^{1/3} \text{ cm}$ (D) $\left(\frac{5}{11}\right)^{1/3} \text{ cm}$
2. A light ray is incident on a transparent sphere of index $= \sqrt{2}$, at an angle of incidence $= 45^\circ$. What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections and then refracts out into air ?
- (A) 270° (B) 240°
- (C) 120° (D) 180°

3. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $2.7 \times 10^{-8} \Omega \text{m}$ and $1.0 \times 10^{-7} \Omega \text{m}$, respectively. The electrical resistance between the two faces P and Q of the composite bar is :-



- (A) $\frac{2475}{64} \mu\Omega$ (B) $\frac{1875}{64} \mu\Omega$
 (C) $\frac{1875}{49} \mu\Omega$ (D) $\frac{2475}{132} \mu\Omega$
4. In the figure shown AB is a rod of length 30 cm and area of cross-section 1.0 cm^2 and thermal conductivity 336 S. I. units. The ends A & B are maintained at temperatures 20°C and 40°C respectively. A point C of this rod is connected to a box D, containing ice at 0°C , through a highly conducting wire of negligible heat capacity. The rate at which ice melts in the box is :-
 [Assume latent heat of fusion for ice $L = 80 \text{ cal/gm}$]



- (A) 84 mg/s (B) 84 g/s
 (C) 20 mg/s (D) 40 mg/s

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

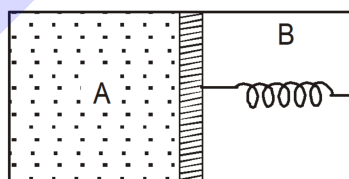
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

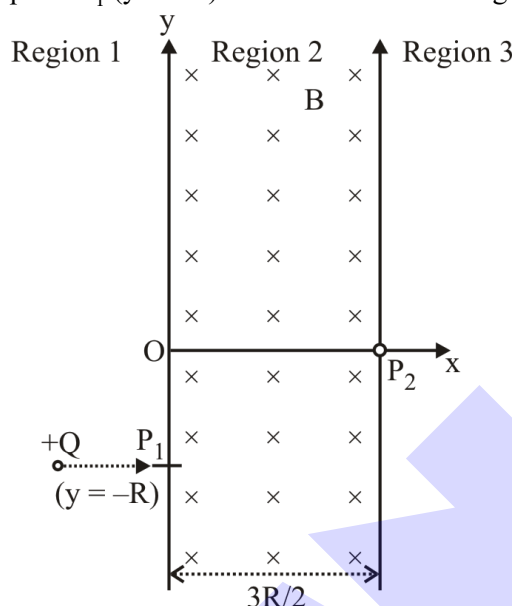
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant k is connected with a massless piston and the wall of the container as shown. Initially spring is unstretched. Gas in chamber A is allowed to expand. Let in equilibrium spring is compressed by x_0 . Then :-

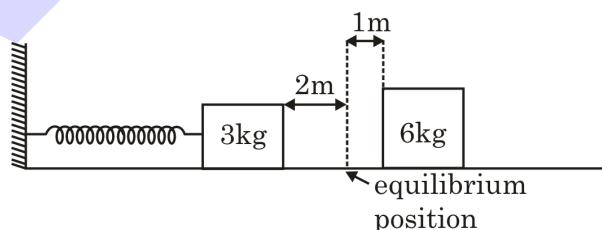


- (A) Pressure of the gas at equilibrium is $\frac{kx_0}{S}$.
- (B) Work done by the gas upto equilibrium is $\frac{1}{2} kx_0^2$.
- (C) Change in internal energy of the gas upto equilibrium is $\frac{1}{2} kx_0^2$.
- (D) At equilibrium temperature of the gas is decreased.

6. A uniform magnetic field B exists in the region between $x = 0$ and $x = \frac{3R}{2}$ (region 2 in the figure) pointing normally into the plane of the paper. A particle with charge $+Q$ and momentum p directed along x -axis enters region 2 from region 1 at point P_1 ($y = -R$). Which of the following option(s) is/are correct?



- (A) For $B = \frac{8}{13} \frac{p}{QR}$, the particle will enter region 3 through the point P_2 on x -axis.
- (B) For $B > \frac{2}{3} \frac{p}{QR}$, the particle will re-enter region 1.
- (C) For a fixed B , particles of same charge Q and same velocity v , the distance between the point P_1 and the point of re-entry into region 1 is inversely proportional to the mass of the particle.
- (D) When the particle re-enters region 1 through the longest possible path in region 2, the magnitude of the change in its linear momentum between point P_1 and the farthest point from y -axis is $p/\sqrt{2}$.
7. Two blocks of masses 3 kg and 6 kg rest on a horizontal smooth surface. The 3 kg block is attached to a spring with a force constant $k = 900 \text{ Nm}^{-1}$ which is compressed 2 m from beyond the equilibrium position. The 6 kg block is at rest at 1 m from mean position. 3 kg mass strikes the 6 kg mass and the two stick together.



- (A) Velocity of the combined masses immediately after the collision is 10 ms^{-1} .
- (B) Velocity of the combined masses immediately after the collision is 5 ms^{-1} .
- (C) Amplitude of the resulting oscillation is $\sqrt{2} \text{ m}$.
- (D) Amplitude of the resulting oscillation is $\sqrt{5}/2 \text{ m}$.

SECTION-I (iii) : (Maximum Marks: 12)

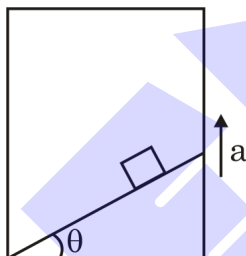
- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. A block of mass m is stationary with respect to a rough wedge as shown in figure. Starting from rest, in time t work done on the block : ($m = 1\text{kg}$, $\theta = 30^\circ$, $a = 2\text{m/s}^2$, $t = 4\text{s}$)



	List-I		List-II
(P)	By gravity	(1)	144 J
(Q)	By normal reaction	(2)	32 J
(R)	By friction	(3)	56 J
(S)	By all the forces	(4)	48 J
		(5)	None

(A) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

(B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

(D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

9. An isolated zinc bead of radius r charged to a negative potential V_0 (assuming potential at infinitely distant points to be zero) placed in free space is being continuously irradiated by ultraviolet light of wavelength λ . Photoelectric threshold wavelength for zinc is λ_0 , speed of light is c , Denoting Planck's constant by h , permittivity of free space by ϵ_0 , charge and mass of an electron by $-e$ and m .

List- I		List-II	
(P)	Maximum speed of a photoelectron immediately after emission is	(1)	$\frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$
(Q)	Maximum speed of a photoelectron at a great distance from the bead is	(2)	$\sqrt{\frac{2hc}{m} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) + \frac{eV_0}{m}}$
(R)	Potential of the bead after prolonged irradiation is	(3)	$\frac{4\pi\epsilon_0 r}{e} \left\{ \frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) + V_0 \right\}$
(S)	Total number of photoelectrons emitted from the bead is	(4)	$\sqrt{\frac{2hc}{m} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)}$
		(5)	None of these

- (A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$
 (B) $P \rightarrow 4; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 3$
 (C) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$
 (D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

10. Two transparent media of refractive indices μ_1 and μ_3 have a solid lens shaped transparent material of refractive index μ_2 between them as shown in figures in **List-II**. A ray traversing these media is also shown in the figures. In **List-I** different relationships between μ_1 , μ_2 and μ_3 are given. Match them to the ray diagrams shown in **List-II**.

List-I		List-II	
(P)	$\mu_1 < \mu_2$	(1)	
(Q)	$\mu_1 > \mu_2$	(2)	
(R)	$\mu_2 = \mu_3$	(3)	
(S)	$\mu_2 > \mu_3$	(4)	
		(5)	

- (A) $P \rightarrow 1,4; Q \rightarrow 2,3; R \rightarrow 3,4; S \rightarrow 1,5$
 (B) $P \rightarrow 1,4,5; Q \rightarrow 1,3,5; R \rightarrow 1,2; S \rightarrow 1,5$
 (C) $P \rightarrow 1,3; Q \rightarrow 2,4,5; R \rightarrow 1,3,5; S \rightarrow 2,4$
 (D) $P \rightarrow 1,3,5; Q \rightarrow 2,4; R \rightarrow 2,3; S \rightarrow 4,5$

11. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in **List-II**. When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltage V_1 and V_2 (indicated in circuits) are related as shown in **List-I**. Match the two list.

	List-I		List-II
(P)	$I \neq 0$, V_1 is proportional to I	(1)	
(Q)	$I \neq 0$, $V_2 > V_1$	(2)	
(R)	$V_1 = 0$, $V_2 = V$	(3)	
(S)	$I \neq 0$, V_2 is proportional to I	(4)	
		(5)	

- (A) $P \rightarrow 3,4,5; Q \rightarrow 2,3,4,5; R \rightarrow 1,2; S \rightarrow 2,3,4,5$
- (B) $P \rightarrow 1,3,5; Q \rightarrow 2,3,4,5; R \rightarrow 3,4; S \rightarrow 2,5$
- (C) $P \rightarrow 3,4,5; Q \rightarrow 2; R \rightarrow 3,5; S \rightarrow 1,4$
- (D) $P \rightarrow 2,4; Q \rightarrow 2,3; R \rightarrow 1,4; S \rightarrow 1,5$

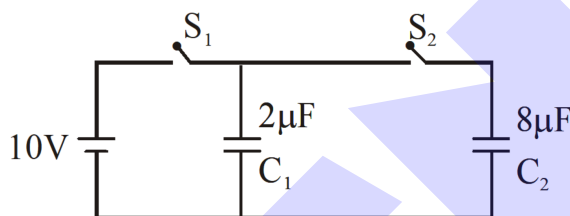
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

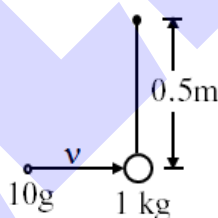
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

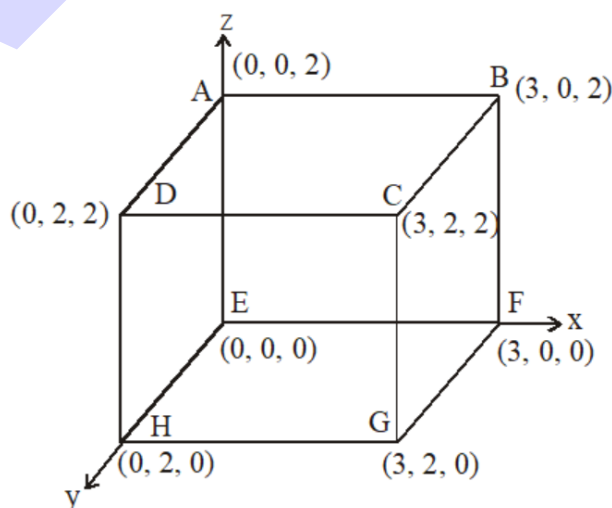
1. A $2\ \mu\text{F}$ capacitor C_1 is first charged to a potential difference of 10 V using a battery. Then the battery is removed and the capacitor is connected to an uncharged capacitor C_2 of $8\ \mu\text{F}$. The charge in C_2 on equilibrium condition is _____ μC . (Round off to the Nearest Integer)



2. A bullet of 10 g, moving with velocity v , collides head-on with the stationary bob of a pendulum and recoils with velocity 100 m/s. The length of the pendulum is 0.5 m and mass of the bob is 1 kg. The minimum value of $v =$ _____ m/s so that the pendulum describes a circle. (Assume the string to be inextensible and $g = 10\ \text{m/s}^2$)



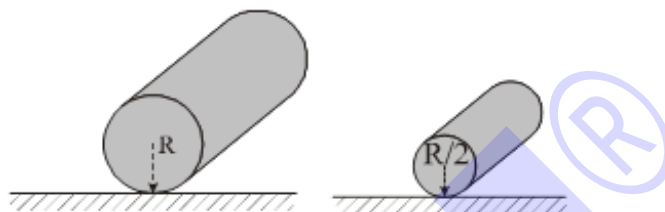
3. An electric field $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j}\ \text{N/C}$ passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_I and ϕ_{II} respectively. The magnitude of difference between $(\phi_I - \phi_{II})$ is (in Nm^2/C) _____.



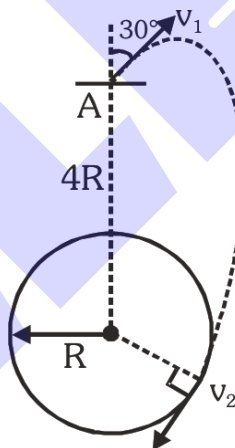
4. A signal of 0.1 kW is transmitted in a cable. The attenuation of cable is -5 dB per km and cable length is 20 km. The power received at receiver is 10^{-x} W. The value of x is _____.

$$[\text{Gain in dB} = 10 \log_{10} \left(\frac{P_0}{P_i} \right)]$$

5. A carpet of mass 'M' made of inextensible material is rolled along its length in the form of a cylinder of radius 'R' and is kept on a rough floor. The carpet starts unrolling without sliding on the floor when a negligibly small push is given to it. The horizontal velocity of the axis of the cylindrical part of the carpet when its radius reduces to $R/2$ is $\sqrt{\frac{x \times 7gR}{3}}$ then find x .



6. A particle is projected from point A, that is at a distance $4R$ from the centre of the Earth, with speed v_1 in a direction making 30° with the line joining the centre of the Earth and point A, as shown. Find the speed v_2 if particle passes grazing the surface of the earth. Consider gravitational interaction only between these two. (use $\frac{GM}{R} = 6.4 \times 10^7 \text{ m}^2/\text{s}^2$) Express your answer in the form $(500\sqrt{2} X) \text{ m/s}$ and write the value of X .



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

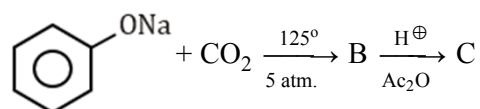
Negative Marks : -1 In all other cases

1. Identify the incorrect statement(s) among the following :
 - (i) Moist ammonia gas can be dried by using anhydrous CaCl_2 .
 - (ii) When metallic copper react with very dilute nitric acid, ammonium nitrite is one of the products.
 - (iii) Phosphine is a weaker base than ammonia because phosphorous atom is less electronegative than nitrogen.
 - (iv) NaH_2PO_2 is an acidic salt and it is named as sodium hypophosphate.

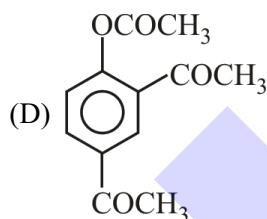
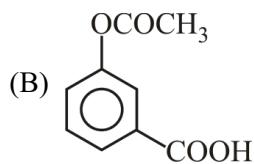
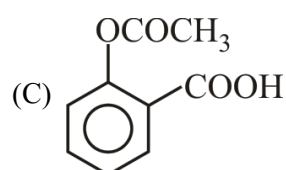
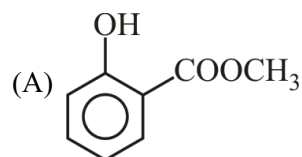
(A) (i) and (iii)
(B) (i) and (iv)
(C) (ii) and (iii)
(D) All of these
2. 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C . The concentration of H^+ at equivalence point is ($K_w = 1 \times 10^{-14}$ at 25°C)

(A) 3.7×10^{-13} M
(B) 3.2×10^{-7} M
(C) 3.2×10^{-2} M
(D) 2.7×10^{-2} M

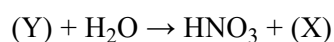
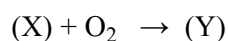
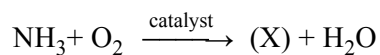
3. Sodium phenoxide when heated with CO_2 under pressure at 125°C yields a product which on acetylation produces C.



The major product C would be :



4. Nitric acid is prepared by ostwald method using given Steps : -



(X) and (Y) are respectively

- (A) NO_2 and NO
 (B) NO and N_2O
 (C) N_2O and NO_2
 (D) NO and NO_2

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

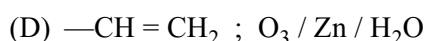
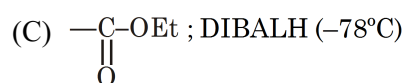
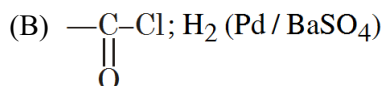
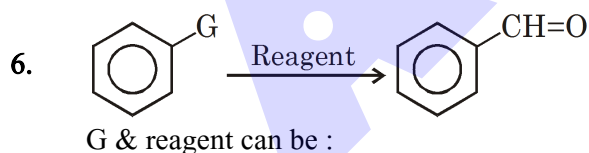
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

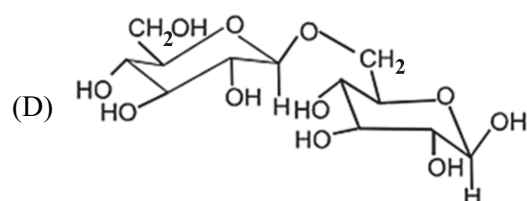
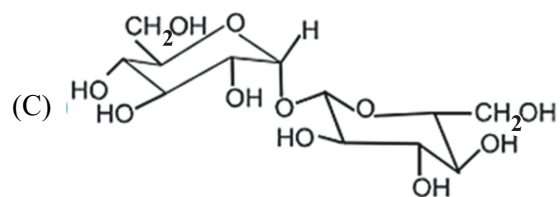
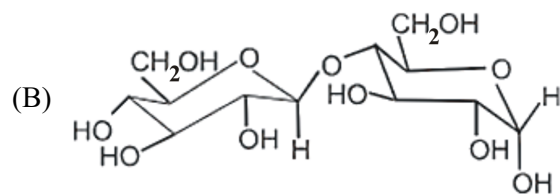
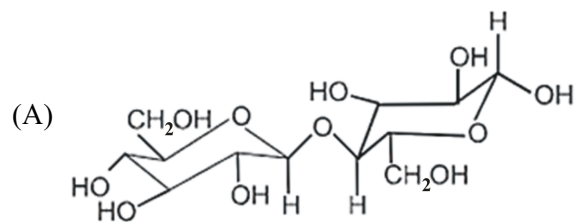
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. The number of correct statements among the following is

- (A) $\text{NF}_3 < \text{NH}_3 < \text{CH}_4 < \text{CO}_2$ (bond angle)
- (B) $\text{S}-\text{H} \cdots \text{O} < \text{O}-\text{H} \cdots \text{S} < \text{F}-\text{H} \cdots \text{O} < \text{F}-\text{H} \cdots \text{F}^-$ (Strength of H-bond)
- (C) $\text{Be}_2\text{H}_4 = \text{C}_2\text{H}_4 < \text{B}_2\text{H}_6$ (max number of H atoms in plane)
- (D) $\text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$ (unpaired electron)



7. Which of the following is/are reducing sugar/s ?



SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. Match the reactions (in the given stoichiometry of the reactants) in List-I with one of their products given in List-II and choose the correct option.

List-I		List-II	
(P)	$P_2O_3 + 3H_2O \rightarrow$	(1)	$P(O)(OCH_3)Cl_2$
(Q)	$P_4 + 3NaOH + 3H_2O \rightarrow$	(2)	H_3PO_3
(R)	$PCl_5 + CH_3COOH \rightarrow$	(3)	PH_3
(S)	$H_3PO_2 + 2H_2O + 4AgNO_3 \rightarrow$	(4)	$POCl_3$
		(5)	H_3PO_4

- (A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 5$ (B) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 4; S \rightarrow 2$
 (C) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$ (D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 5$

9. Match the List :

List-I		List-II	
(P)	Tetragonal and Hexagonal	(1)	are two crystal systems
(Q)	Cubic and Rhombohedral	(2)	$a = b \neq c$
(R)	Monoclinic and Triclinic	(3)	$a \neq b \neq c$
(S)	Cubic and Hexagonal	(4)	$a = b = c$
		(5)	$\alpha = \beta = 90^\circ$

- (A) $P \rightarrow 1,2; Q \rightarrow 1,2,5; R \rightarrow 1,4; S \rightarrow 1,5$ (B) $P \rightarrow 1,5; Q \rightarrow 1,3; R \rightarrow 1,4; S \rightarrow 2,5$
 (C) $P \rightarrow 1,3,5; Q \rightarrow 1,2; R \rightarrow 1,3; S \rightarrow 1,2,5$ (D) $P \rightarrow 1,2,5; Q \rightarrow 1,4; R \rightarrow 1,3; S \rightarrow 1,5$

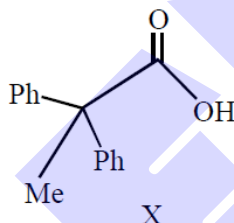
10. $t_{x/y}$ = time in which x/y fraction of reactant converts into product :

List-I		List-II	
(P)	$t_{5/9}$	(1)	Equal to 54 sec. if $t_{1/3}$ is 18 sec. in case of first order reaction.
(Q)	$t_{19/27}$	(2)	Equal to 32 sec. if $t_{1/4}$ is 16 sec. in case of first order reaction.
(R)	$t_{7/8}$	(3)	Equal to 56 sec. if $t_{1/3}$ is 4 sec. in case of second order reaction.
(S)	$t_{7/16}$	(4)	Equal to 30 sec. if $t_{1/3}$ is 18 sec. in case of zero order reaction.
		(5)	Equal to 28 sec. if $t_{1/2}$ is 16 sec. in case of zero order reaction.

- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4,5$ (B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3,5; S \rightarrow 2$
 (C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1,3; S \rightarrow 5$ (D) $P \rightarrow 1,3; Q \rightarrow 2,4; R \rightarrow 3; S \rightarrow 5$

11. The desired product X can be prepared by reacting the major product of the reactions in List-I with one or more appropriate reagents in List-II.

(given, order of migratory aptitude: aryl > alkyl > hydrogen)



List-I		List-II	
(P)	+ H_2SO_4	(1)	$I_2, NaOH$
(Q)	+ HNO_2	(2)	$[Ag(NH_3)_2]OH$
(R)	+ H_2SO_4	(3)	Fehling solution
(S)	+ $AgNO_3$	(4)	$HCHO, NaOH$
		(5)	$NaOBr$

The correct option is

- (A) $P \rightarrow 1; Q \rightarrow 2,3; R \rightarrow 1,4; S \rightarrow 2,4$ (B) $P \rightarrow 1,5; Q \rightarrow 3,4; R \rightarrow 4,5; S \rightarrow 3$
 (C) $P \rightarrow 1,5; Q \rightarrow 1,5; R \rightarrow 1,5; S \rightarrow 1,5$ (D) $P \rightarrow 1,5; Q \rightarrow 2,3; R \rightarrow 1,5; S \rightarrow 2,3$

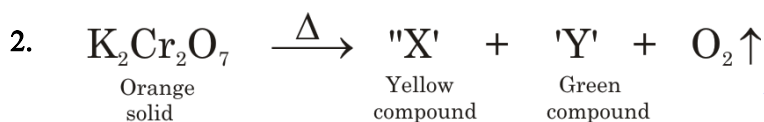
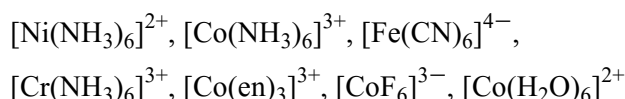
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

1. How many of the following have d^2sp^3 hybridization?



In the given reaction how many statement is **CORRECT**?

- (A) 'Y' is CrO_3
 (B) Aqueous solution of 'X' with conc. H_2SO_4 is converted into orange solution
 (C) When aqueous solution of 'X' is treated with H_2O_2 in acidic medium blue colour compound is obtained
 (D) 'Y' is formed on thermal decomposition of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
3. Consider the following reversible reaction,



The activation energy of the backward reaction exceeds that of the forward reaction by $2RT$ (in J mol^{-1}).

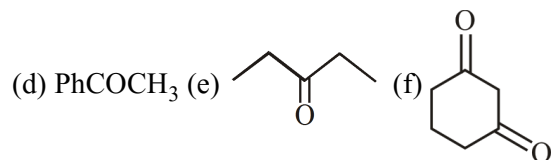
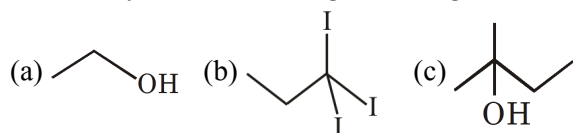
If the pre-exponential factor of the forward reaction is 4 times that of the reverse reaction, the absolute value of ΔG^θ (in J mol^{-1}) for the reaction at 300 K is

(Given : $\ln(2) = 0.7$, $RT = 2500 \text{ J mol}^{-1}$ at 300 K and G is the Gibbs energy)

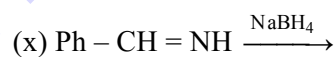
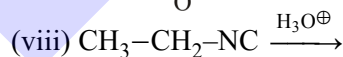
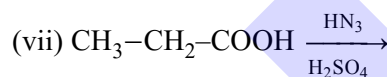
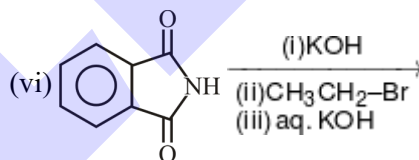
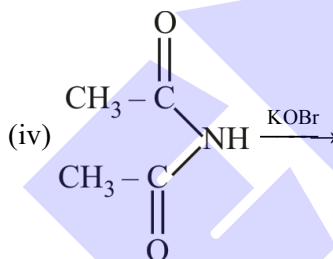
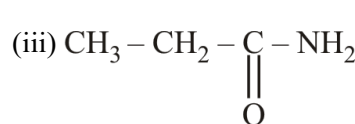
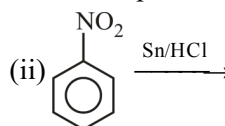
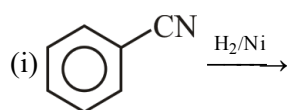
4. Estimate the cell potential (in milli volt) of a Daniel cell having 1M Zn^{++} and originally having 1M Cu^{++} after sufficient NH_3 has been added to the cathode compartment to make NH_3 concentration 2M .
 K_f for $[\text{Cu}(\text{NH}_3)_4]^{2+} = 1 \times 10^{12}$, E° for the reaction,
 $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ is 1.1V .

[Given : $\log 2 = 0.3$, $\frac{2.303 RT}{F} = 0.06$]

5. How many of the following will not give iodoform test :



6. In how many reactions primary amine will be one of the products?



PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. If $\sum_{i=1}^{10} (x_i - 5) = 5$ and $\sum_{i=1}^{10} (x_i - 5)^2 = 125$, variance of $x_1, x_2, \dots, x_{10}$ will be -
- (A) $\frac{9}{4}$ (B) $\frac{25}{4}$
(C) $\frac{49}{4}$ (D) $\frac{81}{4}$
2. Let A and B be two non-empty subsets of set X such that A is not a subset of B, then -
- (A) $A \subseteq B'$ (B) $B \subseteq A'$
(C) A and B are disjoint (D) A and B' are non disjoint
3. Let 'R' be a relation over the set of natural numbers defined by $(p, q) R(r, s) \Rightarrow ps - qr = 0$. Then 'R' is-
- (A) Reflexive only (B) Symmetric only
(C) Transitive only (D) Equivalence
4. Let B be a square matrix such that $|B| = 1$ and $2A + B = B^T - A^T$, then $|A + B|$ is -
- (A) 0 (B) -1
(C) 1 (D) 2

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If n is a positive integer and $(5 + 2\sqrt{6})^{2n+1} = I + f$, where I is an integer and $0 \leq f < 1$, then which of the following is/are true.
- (A) I is an odd integer
- (B) $I + 1$ is divisible by 9
- (C) the integer next above $(5 + 2\sqrt{6})^{2n+1}$ is divisible by 10
- (D) $I - 1 = \frac{f}{1 - f}$
6. If A and B are orthogonal matrices, then which of the following is/are also orthogonal matrix -
- (A) A^2 (B) AB
- (C) A^2B^2 (D) A^TB
7. If extremities of latus rectum of a hyperbola are $(3, 5)$, $(3, -1)$, $(-1, 5)$, $(-1, -1)$ and its transverse axis is parallel to x-axis, then -
- (A) its eccentricity is 2 (B) its eccentricity is greater than 2
- (C) one of its directrix is $x = \frac{1}{2}$ (D) one of its directrix is $x = \frac{3}{2}$

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. India and Australia play a series of 'n' one day matches and probability that India wins a match against Australia is $\frac{1}{2}$.

List-I		List-II	
(P)	If 'n' is not fixed and series ends when any one of the team completes its 4 th win then probability that India wins the series is	(1)	$\frac{4}{2^7}$
(Q)	If n = 7 then probability that India wins atleast three consecutive matches is	(2)	$\frac{47}{2^7}$
(R)	For n = 7; probability that India wins series through consecutive wins is	(3)	$\frac{1}{2}$
(S)	For n = 7; probability that India doesn't win two consecutive matches is	(4)	$\frac{17}{2^6}$
		(5)	$\frac{15}{2^6}$

- (A) P → 4; Q → 2; R → 1; S → 3
 (B) P → 1; Q → 2; R → 3; S → 4
 (C) P → 4; Q → 3; R → 1; S → 2
 (D) P → 3; Q → 2; R → 1; S → 4

9. Match the statement in List-I with given value in List-II

List-I		List-II	
(P)	Number of five digit numbers of the form $d_1d_2d_3d_4d_5$ where $d_i, i = 1, 2, 3, 4, 5$ are digits and satisfying $d_1 < d_2 \leq d_3 < d_4 \leq d_5$, is	(1)	$^{10}C_5$
(Q)	Number of five digit numbers of the form $d_1d_2d_3d_4d_5$ where $d_i, i = 1, 2, 3, 4, 5$ are digits satisfying $d_1 > d_2 \geq d_3 > d_4 > d_5$ is	(2)	$^{11}C_4$
(R)	Pankaj and Prateek play a unique game series in a chess tournament. They decide to play on till one of them wins 5 matches. If each match end only in win or loss. Number of ways in which series can be won by either of them, is	(3)	$^{11}C_6$
(S)	A badminton team has to be selected comprising of 5 students out of 10 students for inter school tournament. Number of ways this can be done if a particular players is to be always included or always excluded from the team, is	(4)	$2 \cdot ^9C_5$
		(5)	$5 \cdot ^9C_5$

(A) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 1, 3; S \rightarrow 1$

(B) $P \rightarrow 1, 3; Q \rightarrow 3, 4; R \rightarrow 2, 3; S \rightarrow 1$

(C) $P \rightarrow 3; Q \rightarrow 1, 3; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 1, 4; S \rightarrow 1, 4$

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	If $\alpha + i\beta$, $\alpha, \beta \in \mathbb{R}$, $\beta \neq 0$ is a root of $z^5 = 1$, then value of $\frac{(\beta^4 - \alpha^4)}{1 - 2\alpha^2}$ is	(1)	0
(Q)	If $z\bar{z} = 1$, then value of $\left 2 + \frac{1}{z}\right ^2 + 2 - z ^2$	(2)	1
(R)	If $\omega = \cos\left(\frac{2\pi}{7}\right) + i \sin \frac{2\pi}{7}$ and $\alpha = \omega + \omega^2 + \omega^4$, $\beta = \omega^3 + \omega^5 + \omega^6$, then value of $2\alpha\beta$ is	(3)	4
(S)	If z_1, z_2, z_3, z_4 represent vertices of a parallelogram (in order), then $z_1 - z_2 + z_3 - z_4$ is	(4)	10
		(5)	6

(A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

11. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Let $(\vec{a} \times \vec{b}) \times \vec{c} = -5\vec{a} + 4\vec{b}$ and $\vec{a} \cdot \vec{b} = 3$ and $\vec{a} \times (\vec{b} \times \vec{c}) = \lambda\vec{b} + \mu\vec{c}$, then $ \lambda + \mu $ is (Where $\vec{a}, \vec{b}, \vec{c}$ are pairwise non-collinear)	(1)	2
(Q)	If the line $y - \sqrt{3}x + 3 = 0$ cuts the parabola $y^2 = x + 2$ at A and B, given that $ (PA)(PB) = \frac{\lambda_1(2 + \sqrt{3})}{\lambda_2}$, then $\frac{\lambda_1^2 + \lambda_2^2}{5}$ is (where P is $(\sqrt{3}, 0)$ and λ_1, λ_2 are relatively prime)	(2)	7
(R)	The line $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies on the plane $2x - 4y + z = 9$, then k is	(3)	9
(S)	Value of $\int_{-\pi/4}^{\pi/4} \frac{x^9 - 3x^7 + x^5 + 1}{\cos^2 x} dx$ is equal to	(4)	5
		(5)	4

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

(D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

1. Number of ways in which 3 boys and 3 girls can be seated on a line where two particular girls do not want to sit adjacent to a particular boy is equal to -
2. A single die is rolled; then 'n' coins are tossed, where n is the number shown on the die. If the probability of exactly four tails is k, then value of 384k is
3. The slope of tangent to the curve $y = xe^{x^2}$ at the point of inflection is
4. Let $f(x) = px^3 + qx^2 + rx$ have local maximum and minimum at $x = 5$ and $x = 1$ respectively. If $\int_{-1}^1 f(x)dx = 6$, then value of $|p + q + r|$ is
5. If the solution of the differential equation $y^2 + \frac{y}{1+x^2} + y' \tan^{-1}x + 2xyy' = y'$ (where $y' = \frac{dy}{dx}$) is given by $y(xy - k + \tan^{-1}x) = c$ (where c is constant of integration) then, k is equal to
6. Let α, β, γ and δ be the roots (real or non real) of equations $x^4 - 3x + 1 = 0$. The value of $\alpha^3 + \beta^3 + \gamma^3 + \delta^3$ is